# Factorising Harder Quadratics

Exam code: 0607

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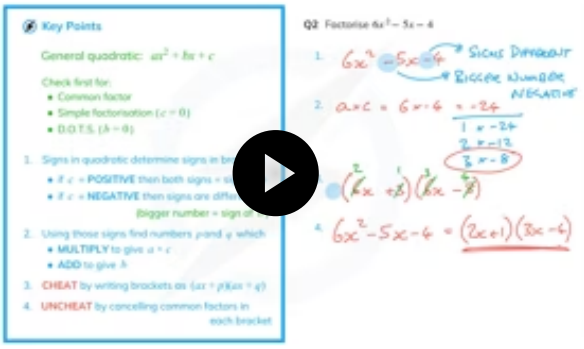

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## Factorising harder quadratics

### How do I factorise a quadratic expression where a ≠ 1 in ax² + bx + c?

#### Method 1: Factorising by grouping

- This is shown most easily through an example: factorising $4x^2 - 25x - 21$
- We need a **pair of numbers** that, for $ax^2 + bx + c$
  - both **multiply** to give *ac*
    - *ac* in this case is 4 × -21 = -84
  - and both **add** to give *b*
    - *b* in this case is -25
  - -28 and +3 satisfy these conditions
  - **Rewrite** the middle term using -28*x* and +3*x*
    - $4x^2 - 28x + 3x - 21$
  - **Group** and fully **factorise** the **first two terms**, using 4*x* as the common factor
  - and **group** and fully **factorise** the **last two terms**, using 3 as the common factor
    - $4x(x - 7) + 3(x - 7)$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

## Method 2: Factorising using a grid

- Use the same example: factorising $4x^2 - 25x - 21$
- We need a **pair of numbers** that for $ax^2 + bx + c$
  - **multiply** to give *ac*
    - *ac* in this case is 4 × -21 = -84
  - and **add** to give *b*
    - *b* in this case is -25
  - -28 and +3 satisfy these conditions
  - Write the quadratic expression in a **grid**
    - (as if you had used a grid to expand the brackets)
    - splitting the middle term up as -28*x* and +3*x* (either order)
  - The grid works by multiplying the row and column headings, to give a product in the boxes in the middle

| | | |
|---|---|---|
| | $4x^2$ | -28*x* |
| | +3*x* | -21 |

- Write a heading for the first row, using 4*x* as the highest common factor of $4x^2$ and -28*x*

| | | |
|---|---|---|
| **4*x*** | $4x^2$ | -28*x* |
| | +3*x* | -21 |

- You can then use this to find the headings for the columns, e.g. "What does 4*x* need to be multiplied by to give $4x^2$?"

| | ***x*** | **–7** |
|---|---|---|
| **4*x*** | $4x^2$ | -28*x* |
| | +3*x* | -21 |

- We can then fill in the remaining row heading using the same idea, e.g. "What does *x* need to be multiplied by to give +3*x*?"

| | ***x*** | **–7** |
|---|---|---|
| **4*x*** | $4x^2$ | -28*x* |
| **+3** | +3*x* | -21 |

## Worked Example

(a) Factorise $6x^2 - 7x - 3$.

**Answer:**

We will factorise by grouping

We need two numbers that:

multiply to 6 × -3 = -18
and sum to -7

-9, and +2

Split the middle term up using these values

$6x^2 + 2x - 9x - 3$

Factorise 2*x* out of the first two terms

$2x(3x + 1) - 9x - 3$

Factorise -3 of out the last two terms

$2x(3x + 1) - 3(3x + 1)$

These have a common factor of (3*x* + 1) which can be factorised out

**$(3x + 1)(2x - 3)$**

(b) Factorise $10x^2 + 9x - 7$.

**Answer:**

We will factorise using a grid

We need two numbers that:

multiply to 10 × -7 = -70
and sum to +9

-5, and +14

Use these values to split the 9*x* term and write in a grid

| | | |
|---|---|---|
| | $10x^2$ | $-5x$ |
| | $+14x$ | $-7$ |

Write a heading using a common factor of 5*x* from the first row

| | | |
|---|---|---|
| **5*x*** | $10x^2$ | $-5x$ |
| | $+14x$ | $-7$ |

Work out the headings for the rows, e.g. "What does 5*x* need to be multiplied by to make $10x^2$?"

| | | |
|---|---|---|
| | $+14x$ | $-7$ |

Repeat for the heading for the remaining row, e.g. "What does $2x$ need to be multiplied by to make $+14x$?"

| | $2x$ | $-1$ |
|---|---|---|
| $5x$ | $10x^2$ | $-5x$ |
| $+7$ | $+14x$ | $-7$ |

Read off the brackets from the column and row headings

$$(2x - 1)(5x + 7)$$

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